



Standard Operating Procedure (SOP)

Use artificial intelligence in support of GxP processes

1937

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Applicability

Division: CH, PH

Area(s): Engineering, IT, Quality

This document is electronically approved (signatures on the last page)

Distribution Sheet for Quality Regulations

Division	CH, PH
Product Lifecycle Phase	CH R&D, Commercial Operation, Manufacturing, PH Biological Development, PH CPD
Region/ Country*	Global
Product Group**	APIs (incl. API starting materials, intermediates), Biotech products, Medical devices, Non-sterile drug products (incl. bulk and combination products), Sterile drug products (incl. bulk and combination products)
Relevance for PH R&D	Shared Relevance PH R&D (Affected)

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Manufactured in or for region / country
Regulation is distributed to each site that handles the product group

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Process Group	Manage GxP computerized system life cycle
Document level	Variant of GSOP 2002 "Manage GxP Computerized System Life Cycle"

1 Scope

This SOP defines the process and quality requirements for using **artificial intelligence in support of GxP processes**. It covers the selection and use of such systems in the GxP environment. The SOP is based on the general framework given in IT-MS-010086 „Compliant use of artificial intelligence”.

This SOP is not based in detail on existing external regulations but describes Bayer requirements to provide a framework in a rapidly changing regulatory environment.

This SOP describes principles that ensure that the use of AI in the context of GxP process covers:

- Data integrity in accordance with ALCOA+
- Traceability and auditability
- Human responsibility (human-in-the-loop)
- Protection of GxP-relevant information

This SOP does **not** cover software that is an integral part of a medical device or is itself classified as a medical device. It also excludes computerized systems used within the PH GLP and GCIS (Global Clinical Imaging Services) areas.

2 Abbreviations

AI	Artificial Intelligence
GenAI	Generative AI
HITL	Human-in-the-Loop

3 Definitions

For generic definitions see SUP 200 “Glossary” and IT-MS-010086 “Compliant Use of Artificial Intelligence (AI)”:

Approved AI systems	AI systems approved by Bayer for internal use that have been released by IT, Information Security, and Data Privacy (LINK). Note: General approval does not constitute validation or release for a specific GxP use case.
Artificial Intelligence (AI)	refers to systems that display intelligent behavior by analyzing their environment and taking actions – with some degree of autonomy – to achieve specific goals
Discriminative AI	Discriminative or Predictive AI has the goal to predict forecasts for future incidents or outcomes based on previous data.
Generative AI (GenAI)	Generative artificial intelligence learns from input training data and generates new data. It is capable of generating, e.g., text, images, videos and have the goal to produce novel and unique material. Examples include: MyGenassist, CoPilot
Human in the Loop (HITL)	Human-in-the-loop refers to the capability for human intervention in every decision cycle of the system In the context of this SOP this means that a qualified human reviews and approves every GxP-relevant AI output or decision before it is implemented, relied upon or released.
Non-approved AI systems	Non-released AI tools (e.g., ChatGPT) that are not allowed to be used for business purposes

4 Process Information

4.1 Purpose and General Guidance

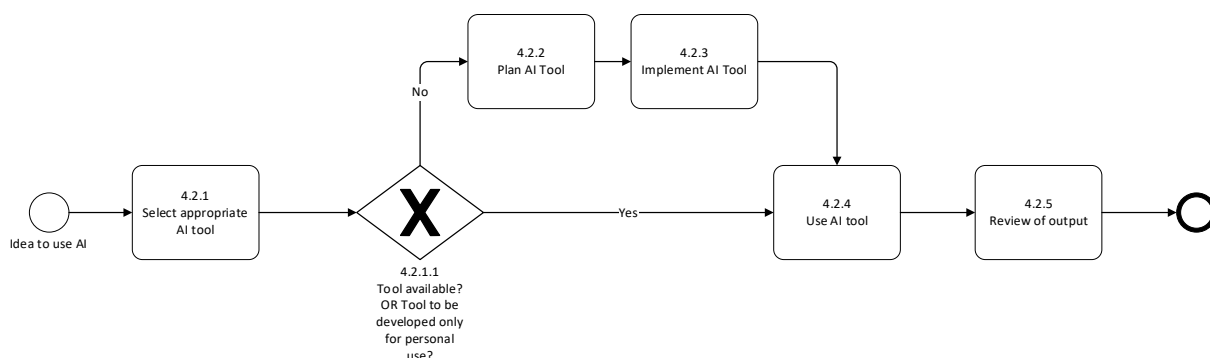
AI tools are used across a broad range of activities, including dedicated systems for defined processes and tools such as agents, prompts, assistants such as MyGenAssist or CoPilot for individual tasks. This SOP covers all such AI tools.

The use of AI in GxP processes, particularly of Generative AI, requires enhanced control. The following rules apply:

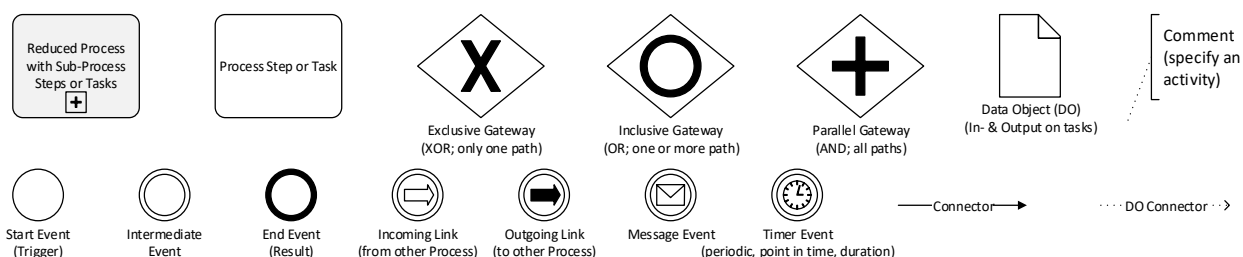
- Generative AI is used only as a supportive tool.
- Generative AI does not autonomously make GxP-relevant decisions.
- Their outputs are not adopted for the documentation of GxP results/assessments without prior review by the respective Subject Matter Experts / persons responsible for the process steps supported by the AI.
- Responsibility for decisions always remains with and is attributable to the user. The final decision lies with the human in-the-loop by reviewing and confirming or approving results delivery by AI.
- The use of AI does not bypass existing GxP processes or controls. This includes also responsibilities as defined in SOPs and training requirements for both the processes supported by AI and necessary technical knowledge when implementing AI tools.

For generic definitions of the roles below, see GSOP 2002 “Manage GxP computerized system life cycle”.

4.2 Process Flow



Legend (Bayer Customized Shapeset):

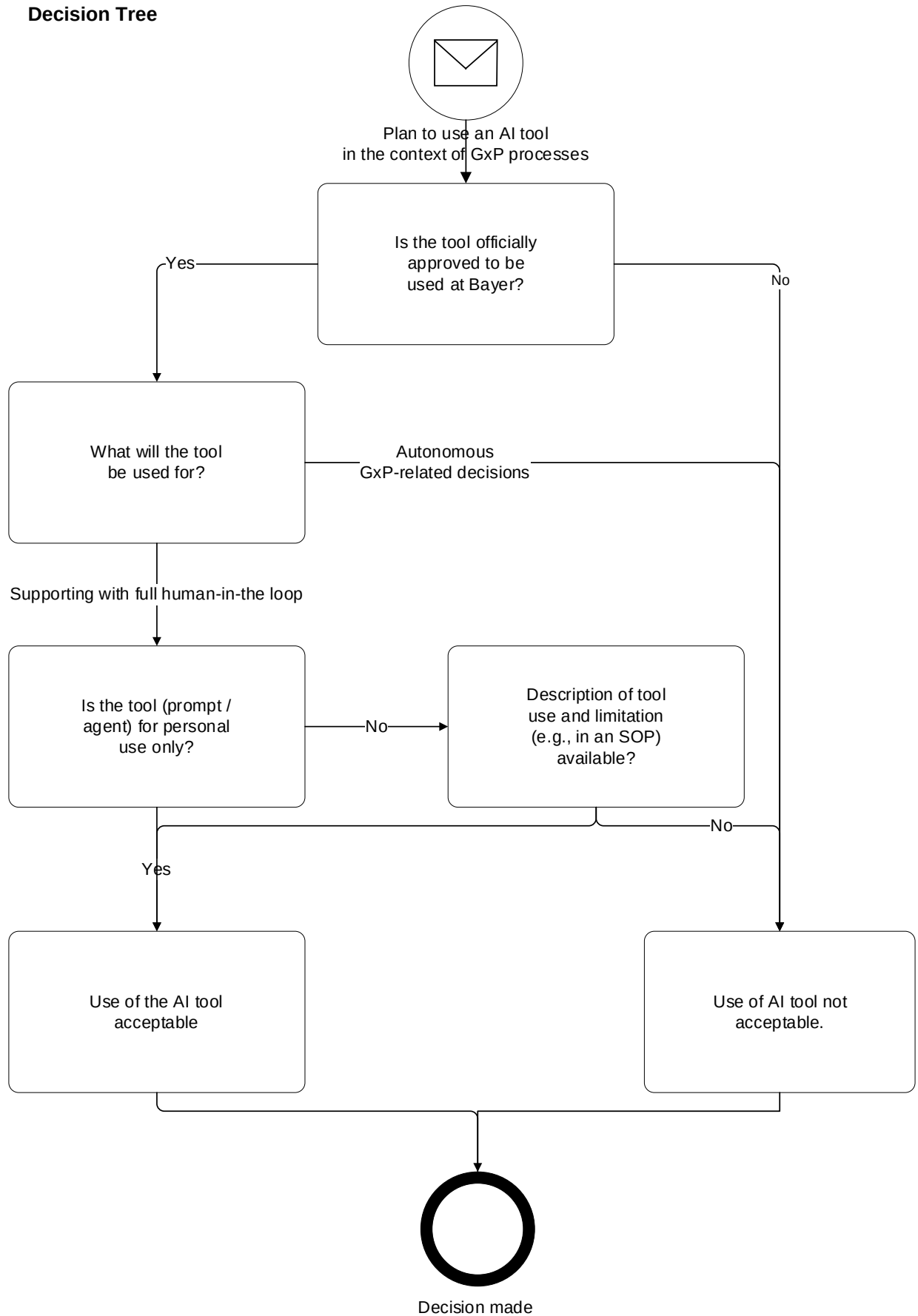


Description including document reference	Assigned Roles
4.2.1 Select appropriate AI tool – Define the use case that you would like to support with AI. Where Generative AI is used in support of GxP processes, use the tool only for supportive activities. For example: • Drafting support for texts where relevant contents has been provided by the user • Language and translation support • Extracting and summarizing data from unstructured sources (e.g. documents) • Structuring/summarizing existing texts • Idea generation (brainstorming) • Support for searching and retrieval – Follow IT-MS-010086 „Compliant use of artificial intelligence” and IT-MS-010113 „AI Compass for Employees” for the general framework, considering also IT Security and Data Privacy considerations. – Evaluate whether there are any data integrity and with that traceability and auditability requirements to be considered for the AI Tool in the context of the supported process. Note that Generative AI does not provide full reproducibility and traceability and therefore is not ALCOA+ compliant. – Where AI builds on existing data (e.g. for searching and retrieval), ensure that the data sources are known. – Verify whether an existing tool, prompt, agent, or system already meets the intended purpose. Prefer existing tools over generating new ones. Contact the system owner or process manager if scope and limitations are unclear. – Use the decision tree in section 4.2.6 for identifying whether AI use is permitted for the intended use case. Annex 1 provides some examples for allowed and not allowed use cases. – Use only internal AI systems approved by Bayer that have been released by IT, Information Security, and Data Privacy (see also overview-of-approved-genai--ai-powered-tools_en_new_2026.pdf) – Evaluate if the tool is appropriate in the context of the intended use case for impact and risks to product quality and patient safety. General approval of a tool does not imply that it is appropriate for a specific use case. – Do not use Generative AI for • approving and releasing GxP-relevant decisions • serving as the sole basis for GxP-relevant process steps.	User

Description including document reference	Assigned Roles
4.2.1.1 AI Tool available OR tool is intended only for personal use? – If an existing tool can be used, then continue with 4.2.4 – If a new tool is needed and it will be only for personal use, i.e. intended to be used by the user himself only, then continue with 4.2.4 – If a new tool is needed and it will be for more than personal use, continue with 4.2.2	
4.2.2 Plan AI tool – When planning to implement a new tool, perform a risk/benefit analysis to evaluate whether this tool will bring business benefits and does not impose additional GxP risks compared to the current process – If the assessment is positive, perform a GxP Determination including description of the system use according to GSOP 2003 “Set up and implement a computerized system in the GxP environment” – In case an AI component is added to an existing computerized system, review the respective GxP determination and perform the implementation in the context of a change to the computerized system. Note: – When implementing an AI tool that is intended to be used by several users, the role concept described in GSOP 2002 “Manage GxP computerized system life cycle” applies.	Process Manager
4.2.3 Implement AI tool – Ensure that GxP decisions are always taken with human review and – where necessary - approval – If the tool is GxP relevant, follow GOI 1861 “Manage GxP computerized life cycle for systems using artificial intelligence / machine learning” for validation. – For AI tools that are supporting a GxP process: • Ensure at a minimum that a tool description including intended use and limitations is available, • Ensure that the procedures for the process supported by the AI tool include a description in which steps and how AI is used, including documentation, where necessary.	Process Manager

Description including document reference	Assigned Roles
4.2.4 Use of the AI tool – Use AI systems in the GxP environment only if: • the user understands the content from a technical/subject-matter perspective • the output is not adopted without review • GxP-relevant decision are not taken relying on AI output only, i.e. AI-generated information is only supportive – In tools for personal use, do not combine supportive tasks in a single instruction or prompt. For example: structuring/summarizing existing texts and drafting support for an assessment of the summarized data within one instruction/prompt	User
4.2.5 Review of the Output – Take full accountability for the results provided by the AI tool – Review every AI output for plausibility, completeness, and correctness during use in a GxP process. Verify also any references or source citations as these might be incorrect /fictitious. – Remember that if the output is incorrect, it is not allowed to point back to the AI as the source for errors. – In case of using an existing tool, review the respective documentation (e.g., SOPs) to understand system limitations and boundaries to identify necessary scope of review.	User

4.2.6 Decision Tree



5 External References

- [1] EMA/CHMP/CVMP/83833/2023 “Reflection paper on the use of Artificial Intelligence (AI) in the medicinal product lifecycle”
- [2] Guiding principles on the use of large language models in regulatory science and for medicines regulatory activities
- [3] EMA / FDA Guiding principles of good AI practice in drug development

6 History of changes

Version	Significant changes with respect to previous version	Effective date
1.0	Initial version	See coversheet

7 Annex 1: Use of AI in the GxP environment – Use case collection

This document provides some practical examples of permissible and not permissible use cases for AI-supported tools in the GxP environment.

7.1 Legend

GREEN	Permitted – supportive use with subject-matter review (HITL) of the AI output
RED	Not permitted – GxP-relevant decision

7.2 Some illustrative use cases

7.2.1 Writing SOPs

Use Case	Requirements
Content is provided by the expert. AI checks redundancies and consistency, adherence to standard formats, formal completeness, compares versions	Author / reviewer need to check proposed changes for correctness (no unintended modifications of meaning).
Drafting support for SOP sections, e.g. based on previous version + some general update information provided.	Author / reviewer need to review the contents carefully to verify that the intended changes are reflected properly in the document and no errors are introduced.
AI alone creates the final version including final subject-matter assessment and decision on the factual correctness.	This is not acceptable. Final versions of SOPs require subject-matter authoring, review (QA), and formal approval. AI may be used only for drafting in support of an expert.

7.2.2 Creating a Risk Assessment

Use Case	Requirements
Risk Assessment is written by an expert, AI checks for formal aspects and text improvements (template, structure, risk descriptions)	Author / reviewer need to check proposed changes for correctness (no unintended modifications of meaning).
Risk Assessment content is proposed by the AI based on a previous version or similar risk assessments.	Author / reviewer need to review the contents carefully to verify that the intended changes are reflected properly in the document and no errors are introduced.
AI uses a URS document to perform a risk assessment without expert input and review.	This is not acceptable. The final assessment of criticality in relation to product quality/ patient safety must be done by the experts and reviewed and approved by QA (and additional approvers in accordance with respective procedures)

7.2.3 Deviation Management

Use Case	Requirements
AI drafts deviation report based on keywords/bullet points provided by the user (descriptions, CAPAs, ...)	Author / reviewer need to check proposed changes for correctness (no unintended modifications of meaning).

Use Case	Requirements
Generate and structure root-cause analysis and outcome with AI (brainstorming) Structure root-cause analysis & generate proposed CAPAs (fishbone, 5-Why, etc.)	AI as a brainstorming partner; hypotheses must be reviewed/assessed by the subject-matter expert.
Generate CAPAs using AI based on the description and root cause analysis.	Review by subject-matter expert necessary to ensure that CAPA actions match the described issue and the root-cause analysis.
Let AI determine the root cause analysis without performing your own analysis	This is not acceptable. Root-cause analysis requires human subject-matter expertise; AI output must not be used as a direct basis for decisions—especially not without your own analysis.

Task: Content Approval Verdict: Approve changes & release Approval to be made Effective	Sigrun Seeger (sigrun.seeger@bayer.com) Approver - Content 17-Sep-2026 10:16:46 GMT+0000
Task: Final Approval Verdict: Approve changes & release Final Approval to be made Effective	Katrin Schnittger (katrin.schnittger@bayer.com) Approver 25-Sep-2026 10:16:59 GMT+0000